

DATASCI 531 – Assignment 3

For this assignment you will interact with an API and extract information contained in JSON format. OpenAlex is a fully open catalog of the global research system, with millions of records from different types of sources. You can read more about it here:

<https://docs.openalex.org>

<https://docs.openalex.org/quickstart-tutorial>

Use these tutorials to answer the following questions:

1. How many works have “artificial intelligence” in the title?
2. Filter years since 2020 and group by the country of the author’s affiliation. Produce a table which sorts the countries by number of publications about artificial intelligence.
3. What are the names of the 10 most highly cited papers on artificial intelligence since 2020? How many citations do they have? What are the authors? What institutions are they affiliated with?
4. How has the number of publications with the word “artificial intelligence” changed year-to-year since 2000? How does this compare to the usage of the word “LLM” or “GPT”? Do the analysis for another keyword of your interest. Plot your results in a graph.

For this you will need to use the works API, which gives allows you to extract information about individual publications, to search by topic, country code, etc. Read the page about how to filter

<https://docs.openalex.org/api-entities/works>

OpenAlex has a series of Python tutorials on how to interact with the API, that you can see here. Use these as inspiration for your own analysis. The tutorial on Japanese publications, while focusing on a different set of questions, could be a useful reference.

<https://docs.openalex.org/how-to-use-the-api/api-overview>

<https://docs.openalex.org/additional-help/tutorials>

For this homework you are expected to process JSON formats, automate the way in which you call the API via flow control,

Deliverable

Submit a Jupyter notebook with you results on Canvas. Include markdown chunks to write comments on your findings and explain your methodology.

When submitting your work imagine that you are sharing this with colleagues at work, who have asked you to explain and justify key decisions and findings of your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues - Code is appropriate to answer question - File is appropriately organized, with different selections and comments clarifying the definition of variables and dataset names.	4 points
Description of methodology - Explain key coding decisions via markdown chunks - Clarify any choices about how to clean data - Shows evidence of conducting quality checks (count sample size of merged datasets, check for the presence/absence of missing values, and/or compute any relevant descriptive statistics)	2 points
Results and Visualization - Write a short summary of the findings. - Correct interpretation of results and easy to follow - Includes graphs to display results from different groups - All graphs look professional (have axis and title labels, suitable color scheme, font sizes, legend, etc.)	4 points